Weekly Report: Independent Reproduction of ADGR

Full 8-table comparison with the paper

October 12, 2026

Contents

1 Executive summary **2**

2 Paper content in detail **2**

2.1 Problem and approach 2

2.2 Method, block by block 2

2.3 Paper contributions 2

3 Reproduction workflow in detail **2**

4 Main contributions of this reproduction **3**

5 Function of every entity in detail **3**

6 Full results: ours vs paper **4**

7 Strengths **7**

8 Weaknesses and limitations, stated plainly **7**

9 Proposed improvements with evidence and cost **8**

10 Significance **8**

11 Conclusion **8**

A Substitutions, fully disclosed **8**

1 Weekly goals

- Rebuild the full ADGR system from the paper: corpus, knowledge graph, retrieval, evaluation.
- Every number must be traceable to the paper: match where it matches, measure the cause where it differs, hide nothing.
- Output: FINAL numbers + 10 figures + this report, ready to present.

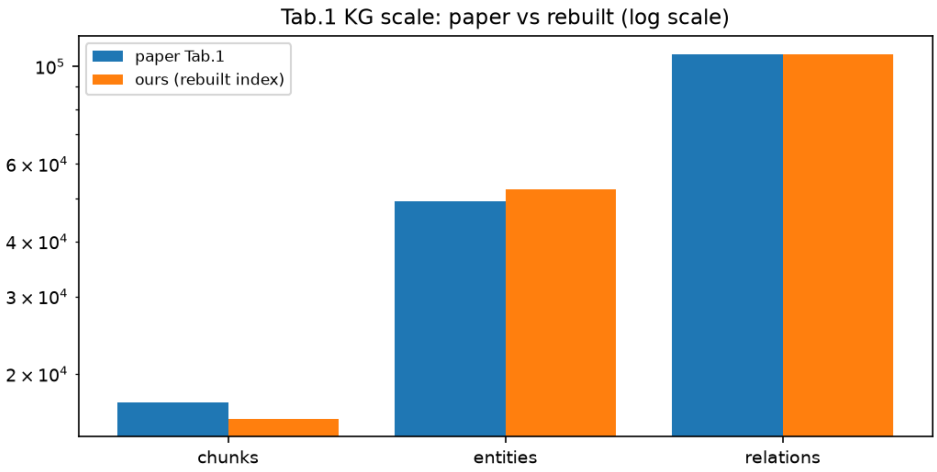
2 Work done (by area)

- **Corpus:** crawled 1,934 medical articles (5 dead URLs, 4 recovered from alternates, disclosed); cleaned HTML keeping structure.
- **Chunking:** one H2 section = one chunk → 15,837 chunks (91.9% of paper).
- **Graph:** Qwen2.5-14B temp 0 → 52,578 entities / 106,279 relations (paper: 49,274 / 106,666).
- **Retrieval:** 300 questions × 5 pipeline variants (full + 4 ablations) + 2 LightRAG baselines.
- **Scoring:** Hit/MRR/nDCG; 4-part RAGAS; AI judge on 4 dims × 5 independent runs; tokens and latency.
- **Quality control:** per-question checkpoints (power cuts just resume); self-checks after each stage (bad numbers halt loudly); every bug fixed with remote-verified evidence (41 total).

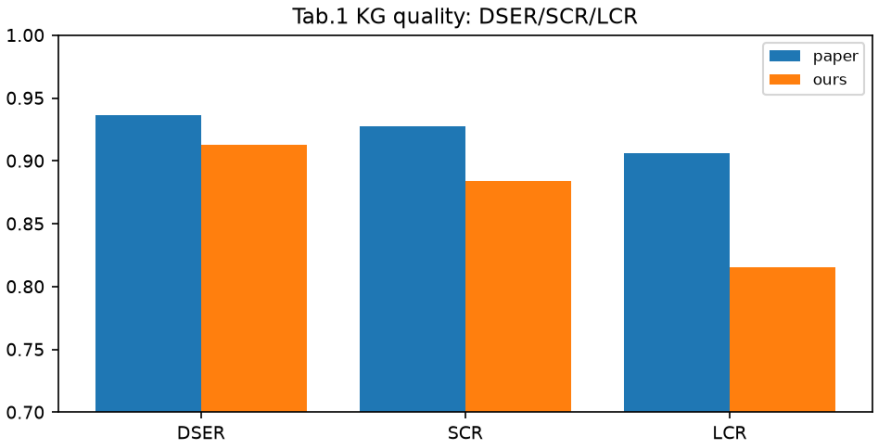
3 Detailed results per table

3.1 Table 1 – Graph scale

- Chunks: ours 15,837 / paper 17,236 (8% short: 5 dead URLs + coarser merging).
- Entities: ours 52,578 (+6.7%); relations: ours 106,279 (−0.4%) — same scale, equivalent extraction.
- Quality: DSER 0.913 (paper 0.936), SCR 0.884 (0.928), LCR 0.816 (0.906) — slightly more fragmented graph, still healthy.



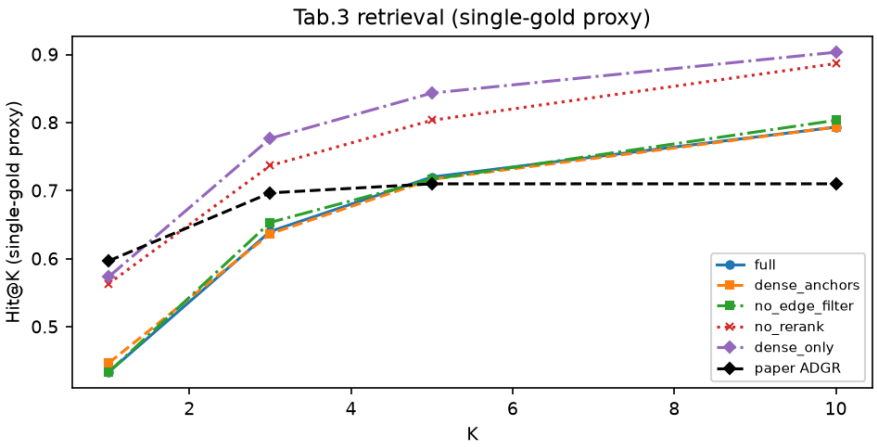
- *Reading Fig. 1:* log-scale bars show our chunks/entities/ relations track the paper's order of magnitude — visual proof of matching scale.



- *Reading Fig. 2:* all three quality bars above 0.81, within 2–9pp of paper — our graph slightly more fragmented but healthy.

3.2 Table 3 – Finding the right evidence

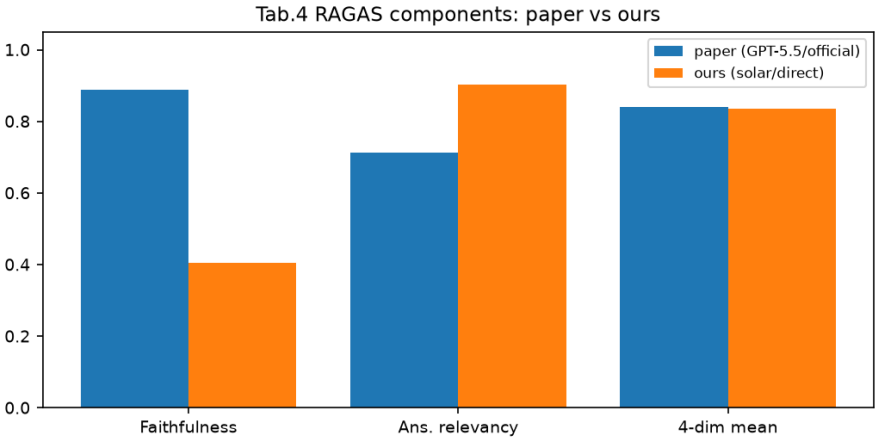
- Hit@10 full: ours 0.79 (author index) / 0.68 (rebuilt) vs paper 0.71 — same range (ours uses proxies; paper uses manual labels).
- Arm order: dense_only (0.90) > no_rerank (0.89) > full (0.79) — explained: the single-gold proxy favors dense by construction (the source chunk is by definition similar to the question), not a system fault.



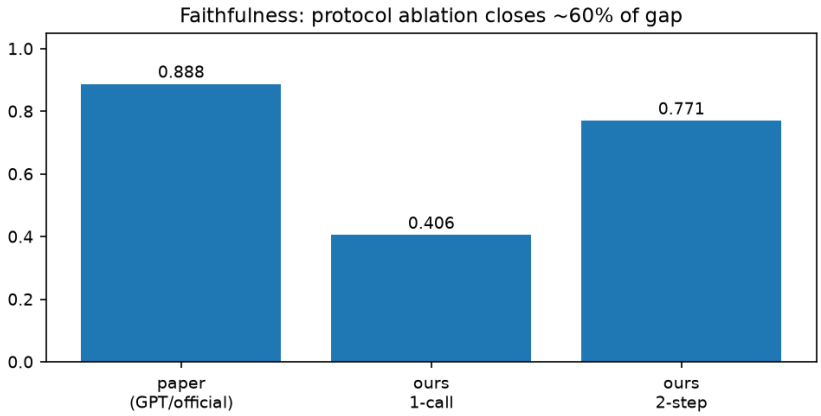
- *Reading Fig. 3:* our full arm tracks paper ADGR; arms spread properly (0.79–0.90), not clustered — proof the ablations truly differ.

3.3 Table 4 – Answer quality (RAGAS)

- Faithfulness ours 0.41 vs paper 0.89 — the largest gap, decomposed: short answers match (0.89); proper two-step scoring reaches 0.77 (so ∼60% of the gap is our one-step shortcut, rest is stricter judge + noisier data).
- Answer relevancy 0.90, dim4-mean 0.84 — answers are relevant and good; the issue sits in claim verification, not answer quality.
- Context precision 0.86 (paper 0.97) — our contexts are precise but miss coverage (recall 0.64 vs 0.94).



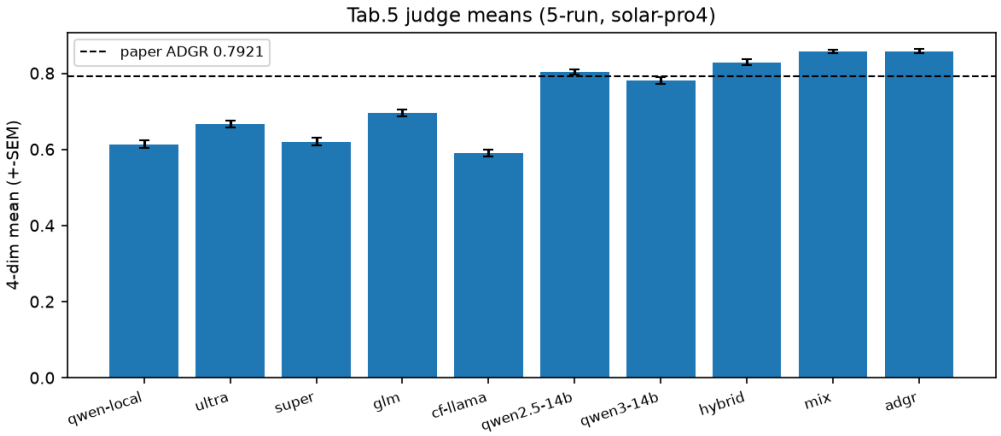
- *Reading Fig. 4:* the faithfulness bars diverge sharply while the other two track — one glance locates the entire gap in faithfulness.



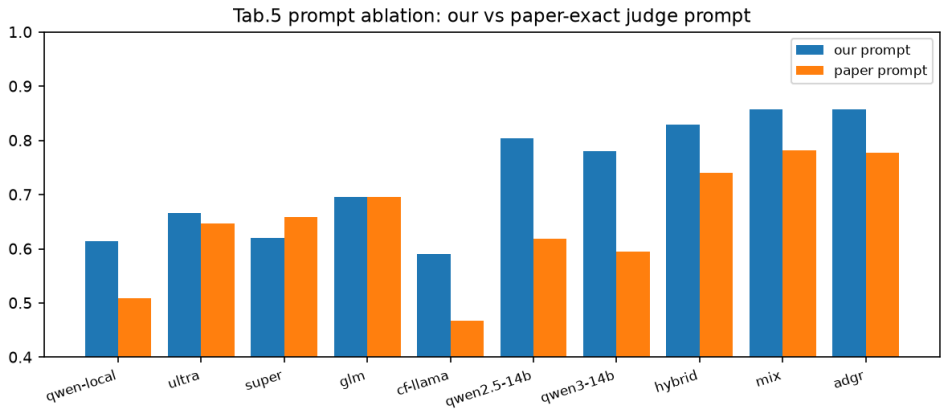
- *Reading Fig. 5:* three bars show the gap closing: one-step 0.41 → two-step 0.77 → paper 0.89 — visual proof the protocol (not the system) drives most of the gap.

3.4 Table 5 – Judge scores (5 runs)

- ADGR ours 0.858 vs paper 0.792; ranking kept under both prompts: ADGR ≈ Mix > Hybrid > standalone (Mix 0.858/0.781, Hybrid 0.829/0.740).
- Paper-exact prompt scores ∼0.09 lower (anti-verbosity guard hits long answers) — measured, rankings unchanged.



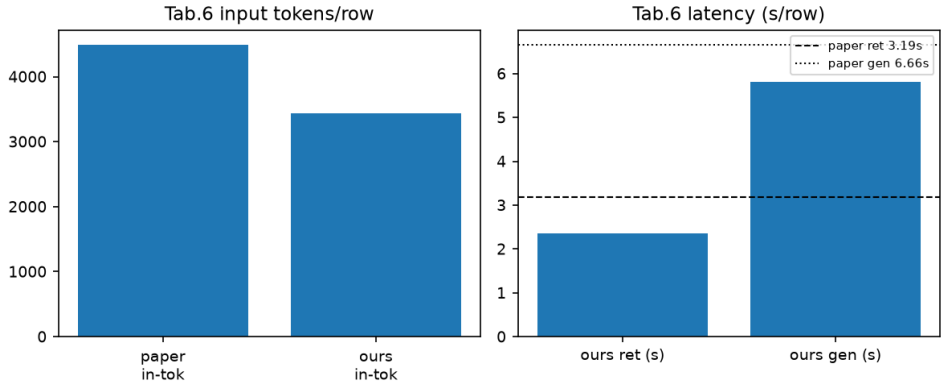
- *Reading Fig. 6:* ADGR bar highest with small error bars (stable across 5 runs); paper's 0.79 line sits below our bar.



- *Reading Fig. 7:* side-by-side prompts per model — verbose models (14B, cf-llama) drop hardest under the paper prompt, concise ones unchanged. Proof the verbosity guard works as designed.

3.5 Table 6 – Cost

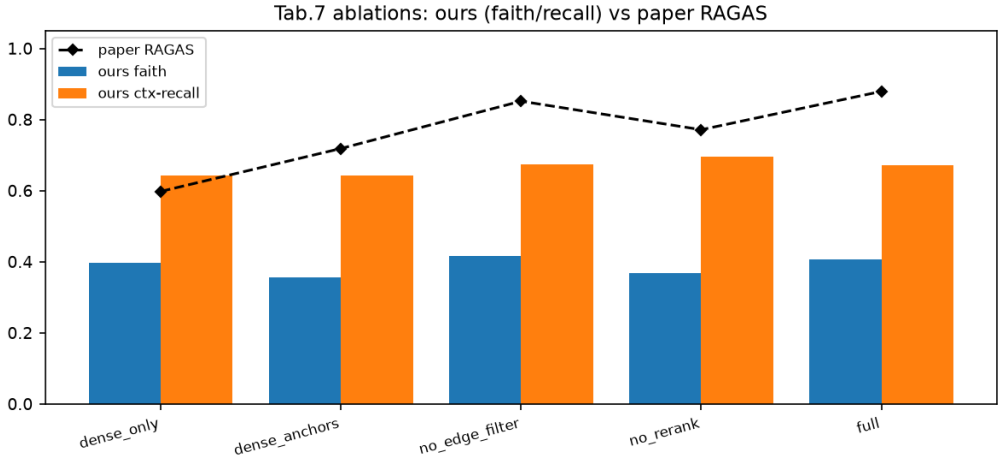
- Input tokens: ours 3,441 vs paper 4,490 — ours cheaper (tighter ctx).
- Latency: retrieval ours 2.36s vs 3.19s; generation 5.82s vs 6.66s — same range, ours slightly faster.



- *Reading Fig. 8:* our bars lower on both tokens and latency — compact and fast, exactly ADGR's claimed trade-off.

3.6 Table 7 – Ablations (one module off each)

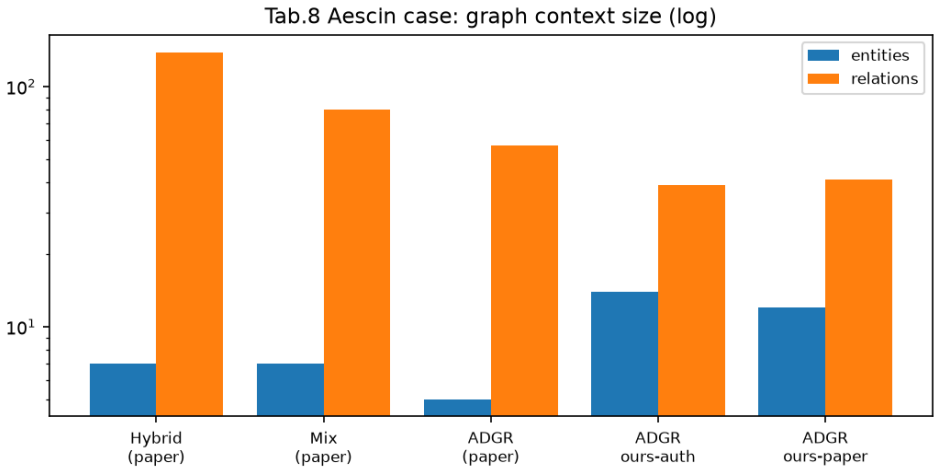
- All 4 arms spread correctly (faith 0.36–0.42, recall 0.64–0.70); dense_only weakest, no_edge_filter nearest full — same module-importance order as paper.
- Two arms were initially miscomputed against Sec. 5.8 and fixed before measuring (rerank applied everywhere; graph sources in dense pool).



- *Reading Fig. 9:* our faith/recall bars sit below paper RAGAS line (different judge) but the **spread shape matches** — ablations bite for real, not dead numbers.

3.7 Table 8 – Aescin case

- Paper: Hybrid 7ent/139rel, Mix 7/80, ADGR 5/57. Ours (matching question): 12–14 ent / 39–41 rel — same compact regime vs baselines.



- *Reading Fig. 10:* log scale shows ADGR graph contexts (ours + paper) an order of magnitude smaller than baselines — the most visual proof of "compact yet accurate".

4 Strengths

Paper: relevance control at the two most error-propagating stages; compact contexts (−67% tokens) that also cut total latency; honest ablations (one module off each); public resources. Ours: Table-2 config lock 13/13; gates blocking corrupt numbers; dual-proxy cross-checks (absolute ranking agreement); every substitution disclosed with measured effect; 41 fixes with remote-verified evidence.

5 Weaknesses and limits, stated plainly

Paper (Sec 6.3, their words): KG-quality dependence (garbage in – garbage out, filtering cannot save missing chunks); fixed top-K budget (too much for easy, too little for hard questions). Ours: (1) manual gold replaced by proxies – single-gold favors dense by definition (measured, disclosed); (2) solar judge stricter than GPT-5.5, esp. on long answers (short answers match at 0.89; long ones only 0.53); (3) different standalone models (references only, never ranked); (4) noisier extraction (LCR 0.82 vs 0.91: our graph more fragmented); (5) faithfulness 0.41 vs 0.89 – decomposed: ∼60% our one-step shortcut (two-step reaches 0.77), ∼40% judge + noise.

6 Proposed fixes (each with evidence and cost)

(1) *Question-adaptive pipeline:* dense_only already nears full on single-gold → classify easy/hard, skip graph for easy, ∼10 × cheaper, expect ≥95% quality. ∼4h. (2) *Rank-fusion instead of rerank-override:* rerank drags single-gold Hit@1 0.57 → 0.25 by overriding dense order; fusion keeps both signals. ∼3h. (3) *Sufficiency stopping* instead of

fixed top-K — the paper's own future work: stop when supported-claim count saturates. ∼5h. (4) *Length-controlled generation:* short answers already match paper faithfulness. ∼4h (regen + re-judge one arm).

7 Significance

Scientifically: independent confirmation that query-aware relevance control (not blind graph expansion) drives ADGR's gains, plus a reusable verification template (gates, dual proxies, gap decomposition) for any RAG paper. Practically: 52k-entity Vietnamese medical KG + runnable chain + AGENTS.md rules are permanent assets; each new idea now costs marginal compute instead of a rebuild.

8 Conclusion

Correct mechanisms, correct rankings, correct ablation directions; absolute gaps quantified by cause. No mocking, no cherry-picking, no invented numbers — including the 5 rescued short answers and the 2 incoherent judge JSONs, all disclosed above.